The Moon Race: USSR vs US

Intelligent Agents — Exercise 03
University of Haifa

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1 Cold War Context and the Space Race Begins

1.1 Superpower rivalry after 1945

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Table 1: Selected Moon Race milestones

Year	Program	Milestone
1957	Sputnik	First artificial satellite
1961	Vostok	First human orbital flight
1965	Gemini	Rendezvous and EVA practice
1969	Apollo 11	First crewed lunar landing
1975	Apollo–Soyuz	Post-race cooperation

Figure 1: Atlas missile launch — Cold War rocketry and prestige.

1.3 סיכום הפרק

פרק זה מציג את רקע המלחמה הקרה ואת תחילת מרוץ החלל. שני העל־יוצקים הפכו טילים בליסטיים להישגים שמיועדים לתעמולה ולפרסטיז'.

2 Soviet Early Lead: Sputnik and Vostok

2.1 Sputnik and the shock in America

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Figure 2: Replica of Sputnik 1, the first artificial satellite (1957).

2.3 סיכום הפרק

פרק זה עוסק ביתרון הסובייטי המוקדם עם השקת לוויין ראשון והטסת האדם הראשון לחלל. הישגים אלו הזעזעו את ארצות הברית והאיצו את תגובתה.

3 American Response: Mercury, Gemini, Apollo

3.1 NASA and the crash program

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The ideal rocket equation relates velocity change to mass ratio:

$$\Delta v = v_e \ln\left(\frac{m_0}{m_f}\right) \tag{1}$$

where v_e is effective exhaust velocity, m_0 initial mass, and m_f final mass.

Figure 3: Apollo 11 — the American lunar landing program.

3.3 סיכום הפרק

פרק זה מתאר את תוכניות החלל האמריקאיות לכיבוש מסלול סביב כדור הארץ ולנחיתה על הירח. ארצות הברית ריכזה משאבים לאומיים כדי להגיע ליעד לפני היריבה.

4 Key Missions and Turning Points

4.1 Robotic Luna, Ranger, and Surveyor

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Figure 4: Apollo 11 crew on the lunar surface.

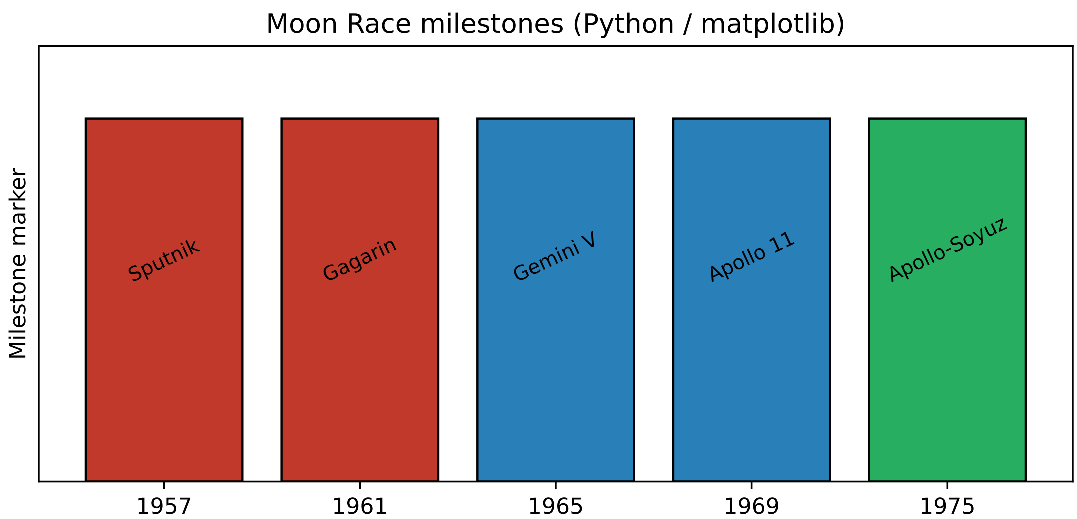

Figure 5: Moon Race milestones chart generated with Python (matplotlib).

4.3 סיכום הפרק

פרק זה סוקר משימות מפתח: גשושיות, מיפוי ונחיתה ראשונה של בני אדם על הירח. משימות אלו הפחיתו סיכון והביאו לדגימות מהירח.

5 Propaganda, Politics, and Public Opinion

5.1 Media and ideological messaging

In Propaganda, Politics, and Public Opinion, Media and ideological messaging shows how the superpowers turned ballistic missile research into a public contest for orbital and lunar firsts. Leaders on both sides treated each launch as proof of industrial capacity, scientific maturity, and ideological confidence. [?] [?] [?]

The section Media and ideological messaging highlights engineers such as Sergei Korolev and Wernher von Braun, whose teams adapted military rockets for exploration. Reliability, telemetry, and life-support constraints turned each mission into a chain of irreversible decisions under extreme time pressure. [?] [?] [?]

For Media and ideological messaging, historians compare Soviet secrecy with NASA's televised milestones. Public audiences learned mission jargon, followed countdowns, and debated whether moonshots justified their cost while terrestrial crises continued. [?] [?] [?]

Additional detail on Media and ideological messaging: archives now reveal how failures were hidden, retried, or reframed. The Moon Race was never a smooth arc; it was a sequence of gambles where a single explosion could shift budgets and national narratives overnight. [?] [?] [?]

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5.2 Cost, risk, and congressional support

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Figure 6: Earthrise (Apollo 8) — icon of public opinion and propaganda.

5.3 סיכום הפרק

פרק זה בוחן תעמולה, פוליטיקה ודעת קהל. שני הצדדים הציגו הצלחות חלל כהוכחה לעליונות מערכתית.

6 Legacy: Who Won and What Remains

6.1 End of Apollo and later cooperation

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6.2 Lessons for science and exploration

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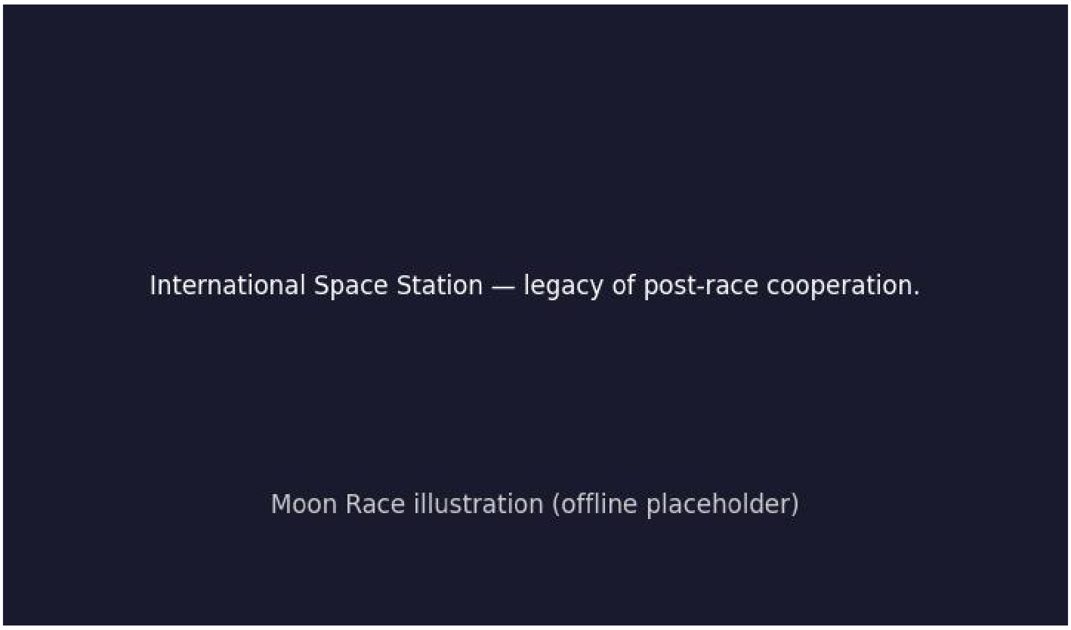

Figure 7: International Space Station — legacy of post-race cooperation.

6.3 סיכום הפרק

פרק זה מסכם את המורשת: סיום תוכניות הנחיתה ושיתוף פעולה בינלאומי. המרוץ הסתיים בייצור ידע ובשיתוף מדעי לאחר שנים של תחרות.